Pi & Planck

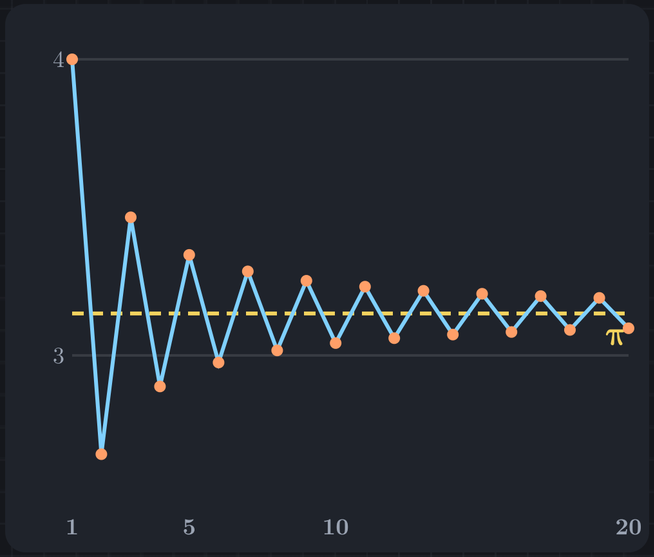
π MATH HISTORY

# Leibniz: Pi From a Simple Sum

Gottfried Wilhelm Leibniz died on 14 November 1716. He gave us calculus notation, binary and a series for π.

## THE LEIBNIZ SERIES

$$\pi / 4 = 1 - 1/3 + 1/5 - 1/7 + 1/9 - \ldots$$




The partial sums swing above and below π and close in slowly.

## THREE LEIBNIZ IDEAS

### Binary, 1703

Every number from just 1 and 0.

1

4

### ∫ and dx, 1675

The notation of calculus we still use.

$$\int f(x)\, dx$$

### HOW SLOW IS IT? (terms for ...)

| | |
|---|---|
| 1 decimal | ≈ 20 |
| 2 decimals | ≈ 200 |
| 3 decimals | ≈ 2 000 |
| 4 decimals | ≈ 20 000 |

## HISTORY

### 14 Nov 1716

Leibniz died in Hanover, aged 70, after a career in mathematics, law, philosophy and diplomacy.

### Found earlier in India

Madhava of the Kerala school knew the series in the 1300s, long before Leibniz. Both deserve the credit.

### Two calculus founders

Newton and Leibniz developed calculus independently; the notation we use today is Leibniz's.

**20** terms to get π right to 1 decimal: 3.1

**20 000** terms to get 4 decimals: 3.1416

**1110** is 14 in binary: 8 + 4 + 2 + 0

**Sources:** MacTutor History of Mathematics, biographies of G. W. Leibniz and Madhava (University of St Andrews) · G. W. Leibniz, Explication de l'arithmétique binaire, Mémoires de l'Académie Royale des Sciences (1703) · Sums and term counts computed for this graphic